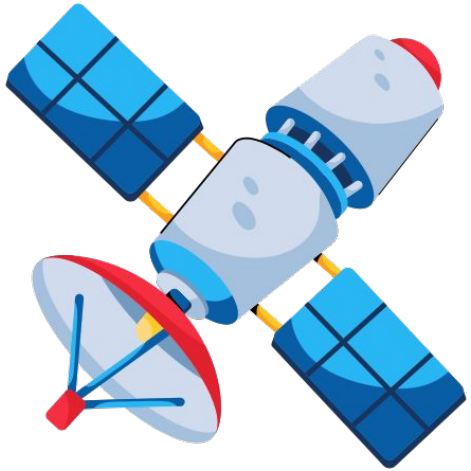


When Satellite and Street Views Disagree:
Cross-View Complementarity in Disaster Damage Assessment

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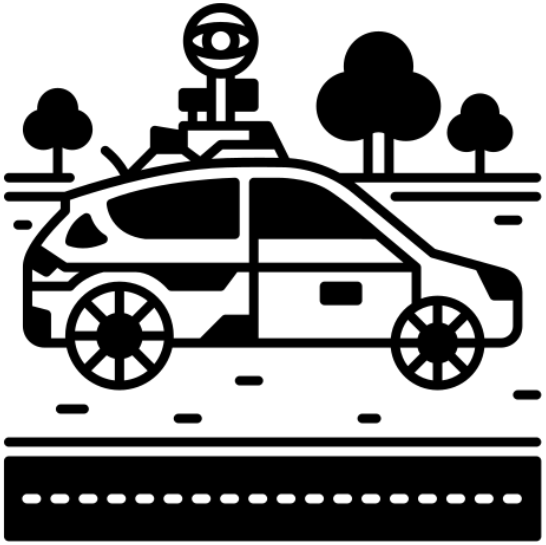
Background

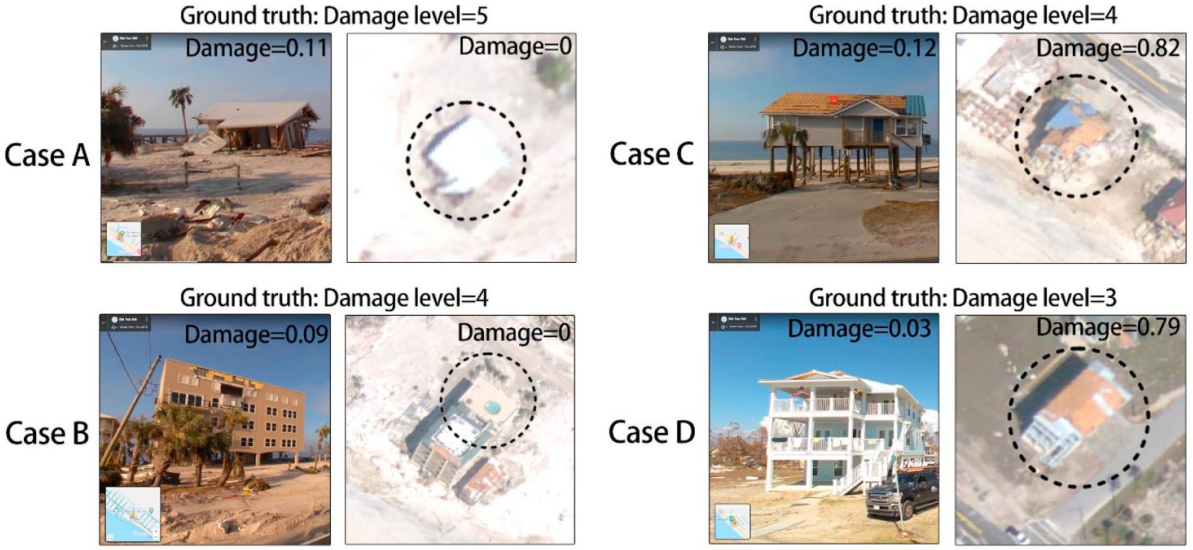




Satellite imagery: rapid, large-scale observation within hours, but it mainly shows roofs

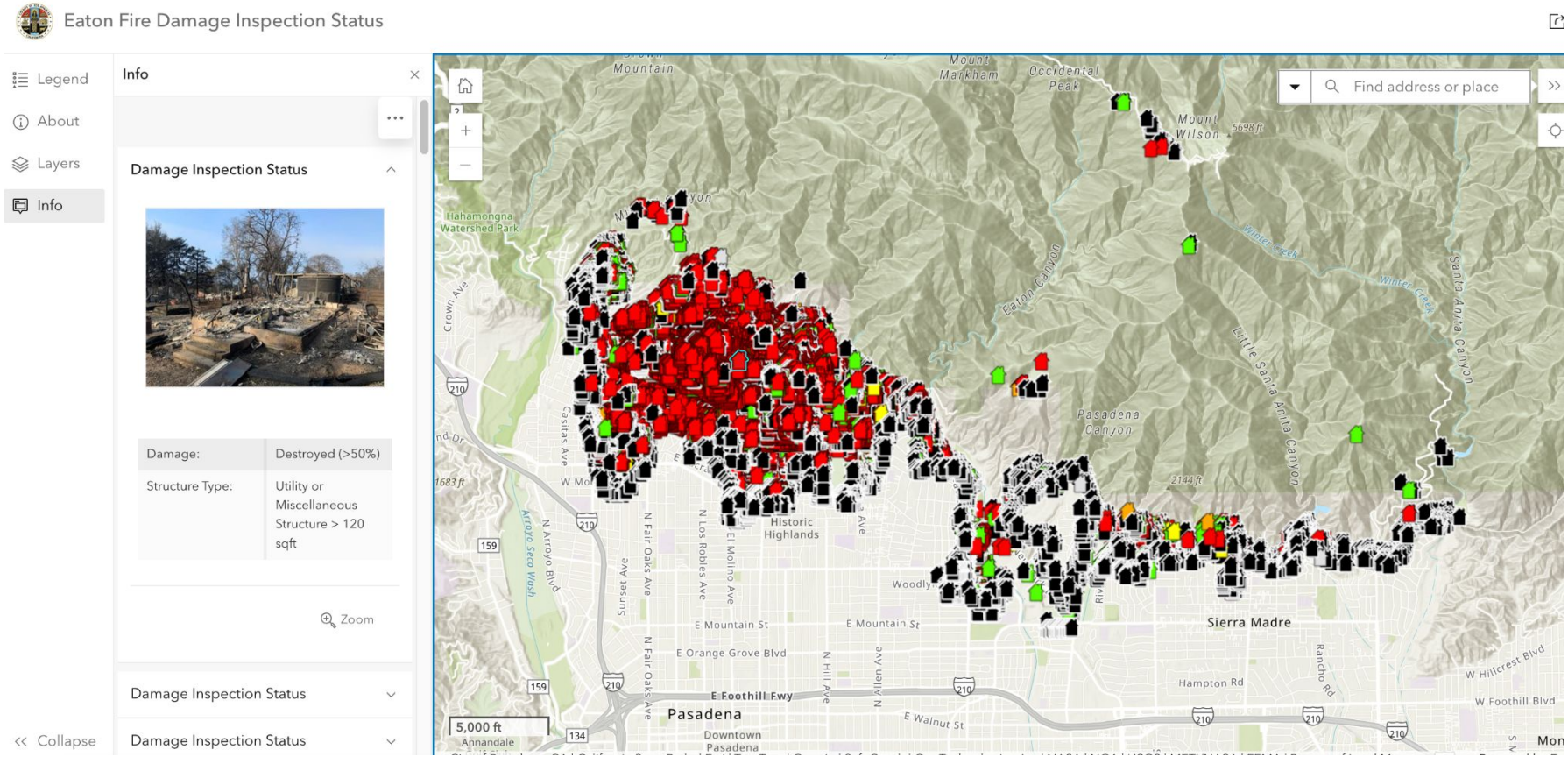
Background





Street-view imagery: facades and debris that overhead sensors cannot fully detect

Background



Eaton wildfire, Altadena CA (Jan 2025): CAL FIRE DINS building-level damage inspections ·
https://lacounty.maps.arcgis.com/apps/instant/sidebar/index.html?appid=2209ecd140d8456291aeba7adbdfd69b

The Problem and Research Questions





Fusion papers report that combining views raises **average accuracy**. But for most buildings the two views agree, so the average hides whether complementary evidence is used.

We mine the **conflict cases** (independently trained single-view models disagree) and ask:

RQ1 How much complementary headroom do fusion methods leave on conflicts?

RQ2 Can a simple, readable gate recover part of it?

RQ3 Does aiming the ground view at the building change the value of fusion?

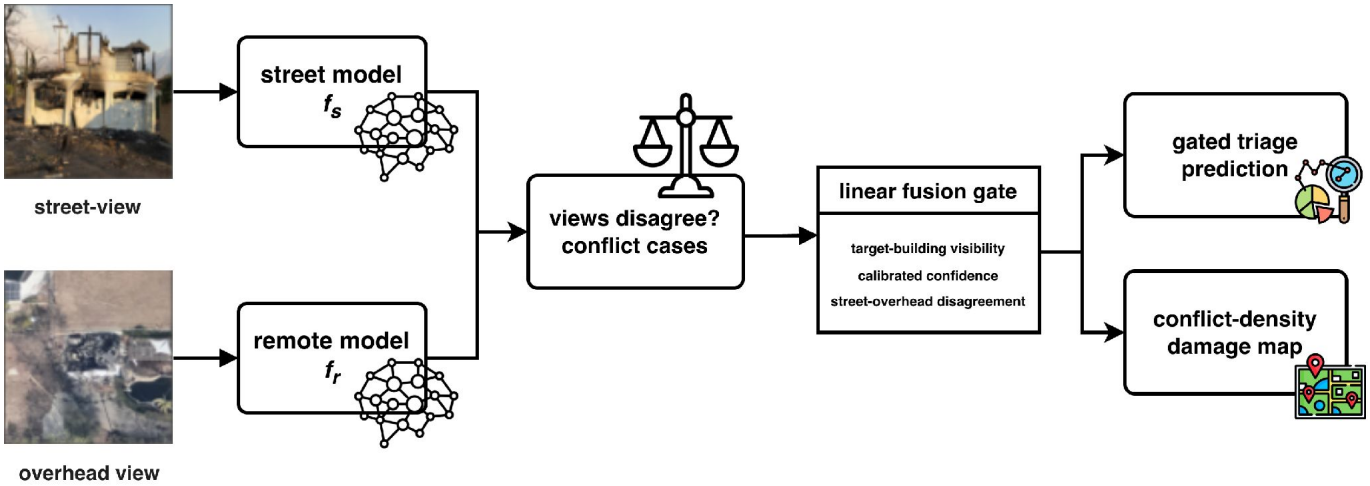
RQ4 Does the spatial density of conflicts track damage, without labels?

Data: Three Paired Cross-View Collections

	Eaton wildfire (Altadena, CA)	Hurricane Ian (Sanibel Island, FL)	Hurricane Milton (Florida)
Event	January 2025 wildfire	September 2022 hurricane	October 2024 hurricane
Ground view	CAL FIRE DINS inspection photos, property-centric (aimed at a building)	Mapillary 360° panoramas (CVIAN pairing, CVDisaster)	360° street-view panoramas, built the same way
Overhead view	Post-event VHR tiles	27 cm post-event satellite imagery	Post-event satellite imagery
Labels	Six-level DINS scale → 3 ordinal classes (no/trace, damaged–repairable, destroyed)	Three-level damage perception	Three-level damage perception
Train / val / test pairs	6.5k / 1,966 / 1,984	3,248 / 573 / 300	1,740 / 307 / 254
Conflict rate (test)	**9.6%**	**33.2%**	**25.7%**

Two hazards × two ground-view regimes: photographs aimed at a building vs. panoramas that are not. Eaton is split by property and tile, so all photographs of one structure fall in one partition.

Method



- **Single-view models** (street, remote) define the conflict cases
- **Oracle** picks the correct single view per conflict → headroom
- **Linear fusion gate** over 11 visibility, confidence and disagreement features
- **Field-of-view intervention**: building-centered vs. random 90° crops
- **Conflict density** per tile as a label-free damage map

Figure 1 of the paper. The implemented gate also mixes in the end-to-end fusion model (omitted in the diagram).

Method: Conflict Cases and the Oracle Gap

Base models, one protocol, five seeds: street_only and remote_only (ResNet-18 + linear head), concat (both encoders + MLP), crossview (learned interaction head = end-to-end fusion reference). Class-balanced CE, 224 px, AdamW, early stopping on val macro-F1.

Conflict subset: C = { x : s(x) ≠ r(x) }, the test samples where the two single-view argmax predictions differ.

Oracle single view: a conflict counts as correct if either view is correct. It measures the headroom of hard view selection between two fixed models; probability mixtures can in principle exceed it.

Oracle-gap closure: *closure(m) = [acc_C(m) − acc_C(best single)] / [acc_C(oracle) − acc_C(best single)]*

Calibration: one temperature per view on validation predictions; every training-free fusion baseline runs on calibrated probabilities (argmax and conflict subset unchanged).



Conflict cases: street view vs. overhead with per-model predictions (GT, street, remote, crossview).

Method: Visibility-Conditioned Fusion Gate

Eleven z-scored features per sample

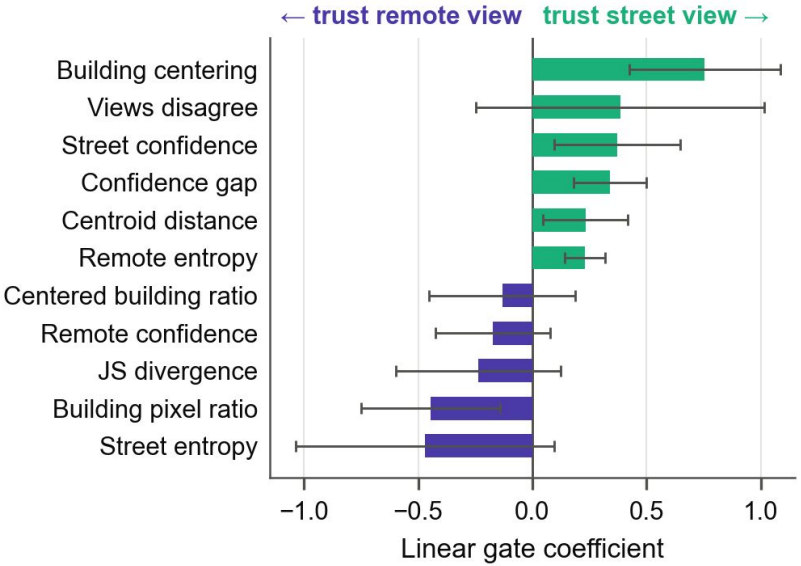
- **Building visibility** (frozen SegFormer-B0, ADE20K): building pixel ratio, ratio inside a centered half-size crop, their difference (centering), normalized centroid distance from image center
- **Prediction cues**: calibrated confidence and entropy per view, confidence gap, Jensen–Shannon divergence, disagreement flag

Mixture of experts with linear logits

$p(y \mid x) = \Sigma_k\, w_k(z)\, p_k(y \mid x)$, $w(z) = \mathrm{softmax}(Wz + b)$, $k \in \{\text{street, remote, crossview}\}$

Fit on the validation split by minimizing mixture NLL. One frozen segmentation pass plus a linear fit: cheap and auditable.

Two-expert variant (street vs. remote) has a single logit, so each coefficient reads as a shift of weight toward one view (figure). A small MLP in place of the linear logits did not help on Eaton.



Two-expert gate coefficients on Eaton (mean ± std, five seeds). Positive = weight toward the street view.

Results: Accuracy, Conflict Accuracy, Closure

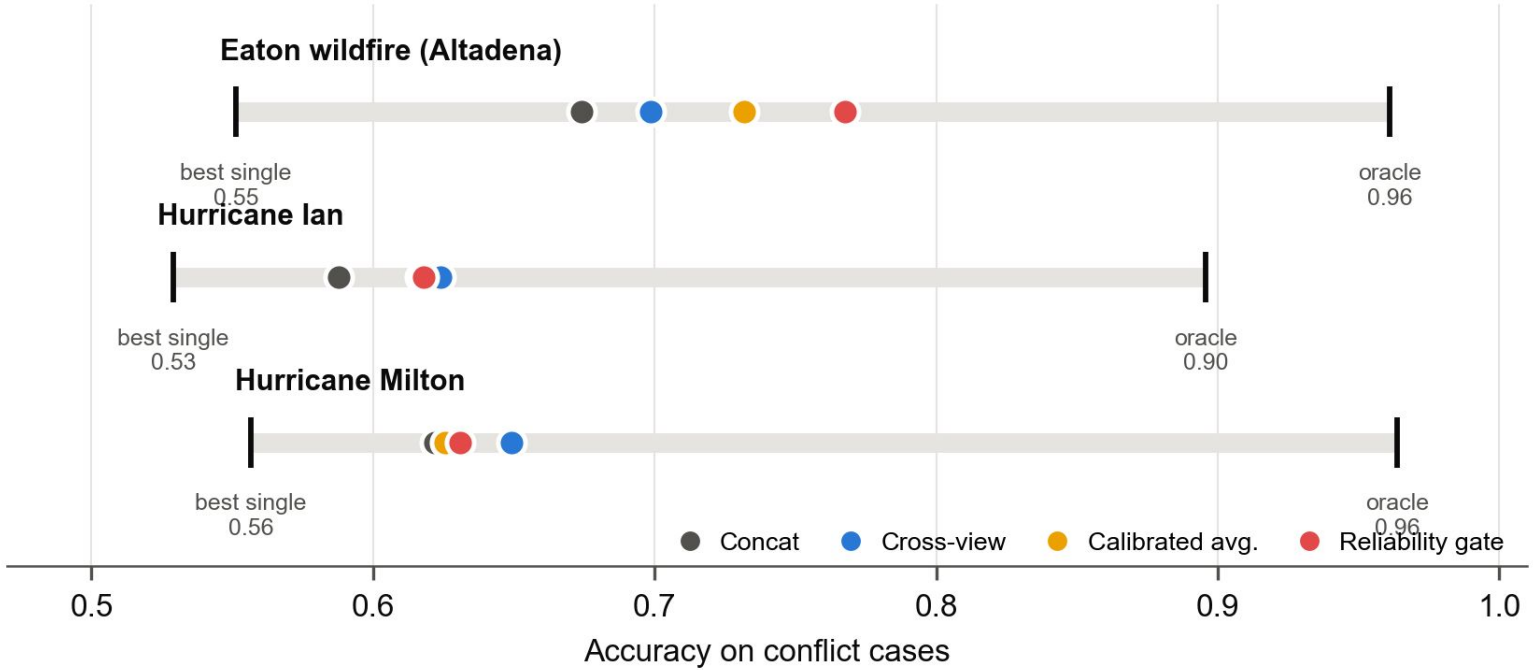
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Test set, five-seed means (accuracy std ≤ 0.03). Conflict accuracy on each seed's conflict subset. Best non-oracle value per column in bold.

	Eaton wildfire (conflict rate 9.6%)			Hurricane Ian (33.2%)			Hurricane Milton (25.7%)		
Method	**Acc.**	**Conflict acc.**	**Closure**	**Acc.**	**Conflict acc.**	**Closure**	**Acc.**	**Conflict acc.**	**Closure**
street_only	0.931	0.486	—	0.703	0.529	—	0.759	0.557	—
remote_only	0.929	0.475	—	0.648	0.367	—	0.721	0.407	—
concat	0.930	0.674	0.40	0.709	0.588	0.16	0.766	0.622	0.16
crossview (end-to-end fusion)	0.940	0.699	0.45	**0.735**	**0.624**	**0.26**	0.776	**0.649**	**0.23**
calibrated probability averaging	0.954	0.732	0.52	0.731	0.617	0.24	0.776	0.626	0.17
fusion gate (linear)	**0.959**	**0.768**	**0.59**	0.731	0.618	0.24	**0.777**	0.631	0.18
oracle single view	—	0.961	1.00	—	0.896	1.00	—	0.964	1.00

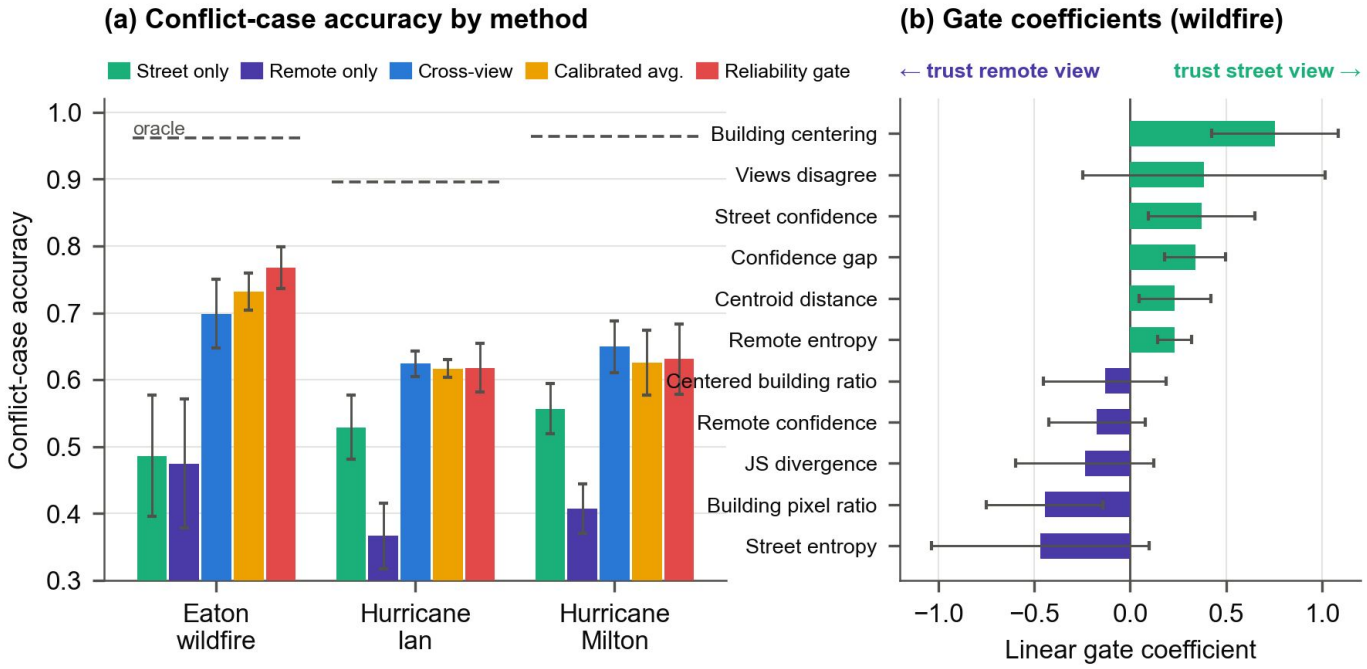
Fusion beats the street model on conflicts everywhere (Eaton +0.246, $p < 10^{-4}$; Ian +0.078, $p = 0.013$; Milton +0.141, $p = 0.006$). **Calibrated averaging is a strong baseline** once single-view models are converged (Eaton 0.954 vs. 0.940 overall). **The oracle dwarfs both:** +0.475 / 0.367 / 0.407 over the better single view and still +0.193 / 0.272 / 0.315 over the best fusion method.

RQ1 · The Oracle Gap Dwarfs Every Fusion Gain



Each bar spans best single view → oracle on conflict cases; dots place each method. Picking the right single view per conflict would add 0.37–0.48 over the better single view; current fusion leaves most view-specific evidence unused.

RQ2 · The Gate Helps on Building-Aimed Photos



Eaton (property-centric)

- +0.072 conflict acc. vs. end-to-end fusion ($p < 10^{-4}$); +0.018 on the full test set
- +0.051 vs. calibrated averaging (95% CI [+0.026, +0.077], p = 0.0001)
- Closure 0.59, highest in Table 1

Ian / Milton (panoramic)

- Indistinguishable from end-to-end fusion (+0.005, p = 0.70; −0.015, p = 0.57)
- Buildings rarely prominent: mean building pixel ratio 0.027 vs. 0.119 on Eaton

Two-expert gate alone (reweight street vs. remote only) reaches 0.734, on par with calibrated averaging and above end-to-end fusion (+0.034, p = 0.047); adding crossview as a third expert lifts it to 0.768. Largest coefficient: building centering (+0.75 ± 0.33); raw building pixel ratio is negative (−0.45 ± 0.31).

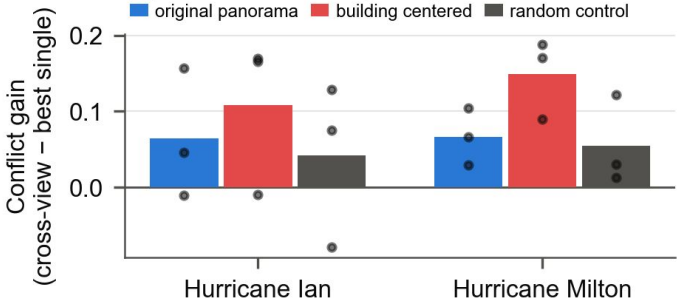
RQ3 · Aiming the Ground View at the Building Raises the Value of Fusion

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(a) One panorama, two geometrically identical 90° crops



(b) Fusion benefit by crop variant





building crop



random crop

Intervention

Each 1024 × 512 panorama → one 256 × 256 window with a 90° field of view, aimed at the SegFormer building azimuth or at a reproducible random azimuth. Same geometry, different pointing.

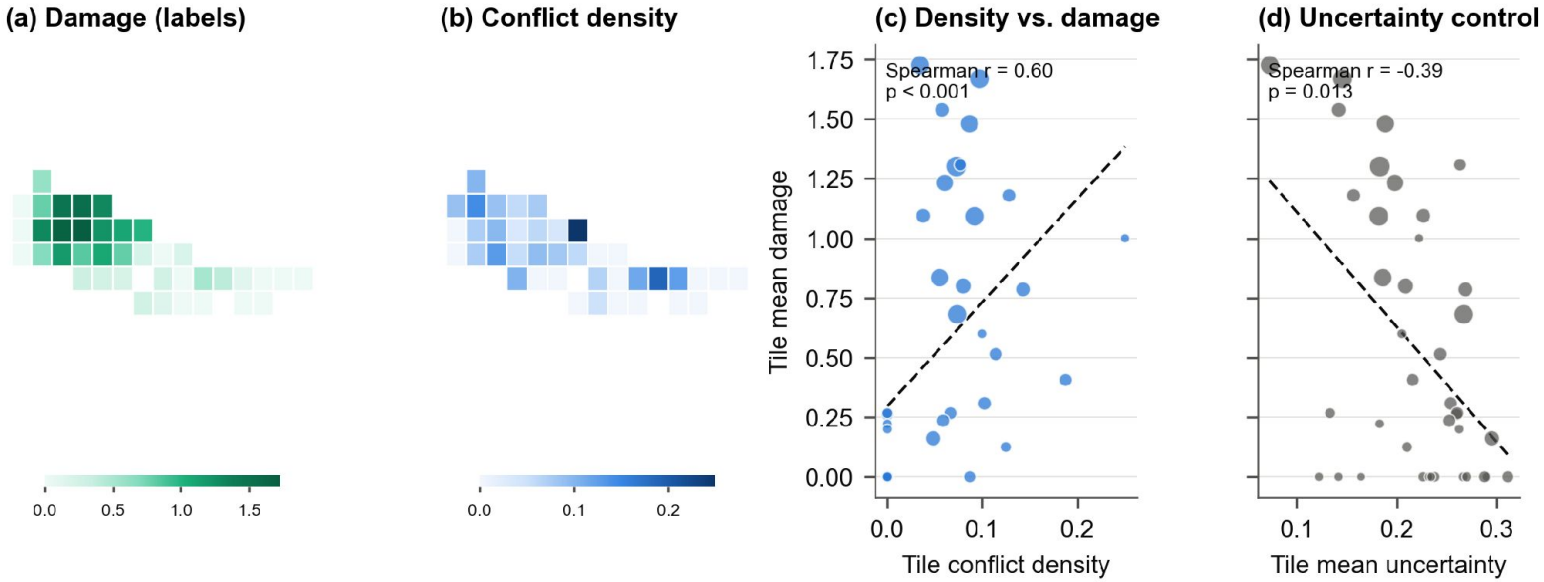
Conflict gain (crossview − best single)

- Ian: 0.064 → 0.109 (random 0.042)
- Milton: 0.067 → 0.149 (random 0.055)
- Building − random positive in 6 / 6 dataset–seed pairs

Cropping lowers street-only accuracy yet raises the benefit of fusion, so the gain is not a stronger street model; the random control argues against a cropping artifact.

Three seeds, three-epoch schedule; remote models unchanged. A fixed-sample test with target-building annotations would be stronger evidence.

RQ4 · Conflict Density Tracks Damage, No Labels



Eaton wildfire, 0.01° grid: tile conflict density (fraction of samples where the two argmax predictions disagree) tracks labeled damage, Spearman r = 0.615 (p = 0.001, 25 tiles), with no labels needed once the models are trained. **The uncertainty control anti-correlates** (r = −0.37 to −0.52): parcels burned to the ground are easy, confident classifications.

Caveats: damage is spatially autocorrelated (Moran's I = 0.27), so nominal p-values overstate the evidence; on Milton the signal is weaker and not significant (r = 0.289, p = 0.14) because better single-view models disagree less.

What Conflict Cases Look Like



Wildfire, property-centric (rows 1–3)

A gutted garage or facade under a roof that still reads intact from above: the street view carries the evidence while the remote model says “no damage” with high confidence (0.01, 0.34, 0.00).

Row 3: street 1 (0.99), remote 0 (0.00), crossview 0 (0.00). End-to-end fusion followed the overhead view and missed. This is the headroom the oracle measures.

Hurricane, panoramic (rows 4–5)

The building is a small patch of the panorama; debris piles along the road drive the street prediction. Visibility cues carry little signal here, which is why the gate cannot beat end-to-end fusion on Ian and Milton.

What a conflict is and is not

We measure prediction complementarity, not which component was damaged. Conflicts may also stem from occlusion, misalignment, or label ambiguity.

Takeaways

- **Conflict cases make complementarity measurable.** The oracle headroom, a restricted choice between two existing models, exceeds every fusion gain we observed: current fusion leaves most view-specific evidence unused.
- **Which view carries the evidence depends on hazard and capture setting.** On building-aimed wildfire photographs a cheap, auditable gate helps: weight the street view when it is confident and the building is centered. On hurricane panoramas calibrated averaging or end-to-end fusion suffices, and framing photographs on structures raises the value of fusion.
- **Spatially aggregated disagreement is a label-free damage indicator** (Eaton r = 0.615), and once single-view models are converged and calibrated, probability averaging is as strong as end-to-end fusion.
- *Mining conflicts turns "does fusion help?" into "which view carries the evidence, where, and why?"*

Code · data · paper

Yifan Yang and Lei Zou. 2026. When Satellite and Street Views Disagree: Cross-View Complementarity in Disaster Damage Assessment. In *The 5th ACM SIGSPATIAL International Workshop on Searching and Mining Large Collections of Geospatial Data (GeoSearch '26)*, November 03–06, 2026, Riverside, CA, USA. ACM, New York, NY, USA, 4 pages. https://doi.org/10.1145/3849732.3857333

Code, protocols and results: https://github.com/rayford295/CrossViewGate · arXiv:2610.04327



Q&A

Welcome any comments/questions

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github.com/rayford295/CrossViewGate

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